

# Piyush Hinduja

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## OBJECTIVE

Data-Driven Machine Learning Engineer with expertise in NLP, LLMs, GenAI, Deep Learning, and scalable data processing. Skilled in architecting high-performance predictive models and leveraging AI to extract actionable insights. Passionate about implementing results-driven business intelligence and enhancing customer-experience through cutting-edge ML solutions.

## EDUCATION

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| <b>Master of Science in Computer Science</b><br><i>University of Utah</i>             | GPA: <b>3.9/4.0</b><br>2023 – 2025  |
| <b>Bachelor of Engineering in Computer Engineering</b><br><i>University of Mumbai</i> | GPA: <b>3.85/4.0</b><br>2019 – 2023 |

## SKILLS

**Languages:** *Expert (5+ yrs):* Python, Java, JavaScript *Intermediate (3+ yrs):* C, C++ *Beginner (1+ yrs):* Kotlin, SQL, Shell  
**Machine Learning:** LLMs, PyTorch, TensorFlow, Scikit-Learn, BERT models, EasyOCR, HuggingFace, Langchain  
**Data Processing:** RESTful APIs, Pandas, NumPy, PySpark, Databricks, OCR Pipelines, Matplotlib, Seaborn  
**Applications and Databases:** Streamlit, Bash, Preswald, MongoDB, MySQL, Amazon RDS, Firebase, ChromaDB  
**Cloud & Miscellaneous:** AWS (EC2, S3, RDS), GCP, Azure, Docker, Kubernetes, CloudLab, Git, CI/CD, Agile/SCRUM, JIRA

## WORK EXPERIENCE

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| <b>Data Analyst</b><br><i>University of Utah</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | March 2024 – Present<br><i>Salt Lake City, Utah</i>  |
| <ul style="list-style-type: none"><li>Implemented advanced NLP analysis using <b>encoder-only model FinBERT, PyTorch, and Transformers</b> for financial document processing, <b>improving the prediction accuracy from 66% to 94.13%</b></li><li>Reduced per-filing <b>processing time by 75% using multi-cores and multi-thread environments</b> to parallelize file processing, API requests, and text cleaning operations</li><li>Developed robust <b>prompt engineering</b> solutions to integrate LLMs &amp; GenAIs (OpenAI, Anthropic, and Gemini) for automated financial text analysis, comparing AI-generated insights against traditional algorithms</li><li>Optimized financial data processing workflows, achieving <b>faster execution and lower memory consumption</b> through optimized data structures and parallelized computations</li></ul> |                                                      |
| <b>Machine Learning Researcher</b><br><i>FLUX Research Group</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | August 2024 – Present<br><i>Salt Lake City, Utah</i> |
| <ul style="list-style-type: none"><li>Engineered automated deployment system using <b>Python and Bash to streamline NVIDIA/CUDA toolkit</b> integration on high-performance computing clusters, cutting setup time by <b>90% for 150+ researchers</b></li><li>Implementing <b>time-series forecasting model</b> to analyze the campus's wireless spectrum data, optimizing power distribution across 100+ routing devices on campus</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                      |
| <b>Full Stack Developer Intern</b><br><i>Exposys Data Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | June 2021 – July 2021<br><i>Bangalore, India</i>     |
| <ul style="list-style-type: none"><li>Spearheaded the launch of a tourism project, integrating key features to <b>enhance user experience</b> and deliver valuable information about 50+ Indian travel destinations</li><li>Utilized <b>MERN stack</b> for an efficient frontend and backend integration, creating a responsive, user-friendly interface and enabling content management</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                      |

## PROJECTS

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| <b>SEC Comment Letter Analysis   LLMs, OCR, Claude API</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |
| <ul style="list-style-type: none"><li>Engineered a rolling-window algorithm to process and bundle SEC comment letter conversations, analyzing data from <b>50,000+ companies</b> via the EDGAR database</li><li>Designed an <b>OCR pipeline using the EasyOCR library</b> to extract text from PDFs with <b>95% accuracy</b>, processing 2-page documents in under 2 seconds</li><li>Constructed prompt-based labeling system using <b>Claude API</b> to classify letter content with <b>98.25% accuracy</b>, integrating metadata for contextual inference</li></ul>                                                   |                             |
| <b>Road Damage Detection   Object-Detection, YOLOv5, FasterRCNN</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">GitHub Link</a> |
| <ul style="list-style-type: none"><li>Collaborated with a team of four to develop a road damage detection system using <b>YOLOv5 and FasterRCNN</b>, integrating a user-friendly web interface (deployed via <b>Streamlit</b>), leveraging the RDD-2020 dataset</li><li>Achieved model accuracies of <b>52.67% with YOLOv5 and 44.45% with FasterRCNN</b> by training and fine-tuning on different hyper-parameters</li><li>The system plays a vital role, delivering a <b>70% improvement in road damage detection accuracy</b> over manual methods, helping lower the frequency of accidents and fatalities</li></ul> |                             |
| <b>Dependency Parser   PyTorch, GloVe Embeddings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">GitHub Link</a> |
| <ul style="list-style-type: none"><li>Architected an end-to-end <b>Transition-Based Dependency Parser in PyTorch</b>, implementing a multi-class classifier for predicting parsing actions based on stack and buffer states</li><li>Designed modular pipeline incorporating data preprocessing, model architecture, training workflow, and evaluation metrics</li><li>Integrated <b>GloVe word embeddings</b> for enhanced semantic representation, contributing to <b>UAS of 0.76 and LAS of 0.705</b></li></ul>                                                                                                       |                             |
| <b>Language Modeling   NLP, RNN, N-gram, PyTorch</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">GitHub Link</a> |
| <ul style="list-style-type: none"><li>Developed <b>character-level LSTM model</b> in PyTorch to predict character sequences based on <b>50-character context windows</b></li><li>Implemented teacher forcing during training to prevent error propagation, using ground truth characters instead of predictions for subsequent timesteps</li><li>Achieved <b>perplexity of 9.43</b> through systematic model architecture optimization and training methodology</li></ul>                                                                                                                                               |                             |